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Give me the answer

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Time Limit: 1 s Memory Limit: 32 MB
Submission: 102 AC: 39 Score: 99.16

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Description

Farmer John commanded his cows to search for different sets of numbers that sum to a given number. The cows use only numbers that are an integer power of 2. Here are the possible sets of numbers that sum to 7:

- 1) 1+1+1+1+1+1+1
- 2) 1+1+1+1+1+2
- 3) 1+1+1+2+2
- 4) 1+1+1+4
- 5) 1+2+2+2
- 6) 1+2+4

Help FJ count all possible representations for a given integer N ($1 \leq N \leq 1,000,000$)

Input

The first line of the input contains the number of test cases in the file. And the first line of each case contains one integer numbers n

Output

For each test case, output a line with the ans % 1000000000.

Samples

input	Copy
1 7	
output	Copy
6	

Source

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Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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Server Time: 2025-8-26 14:59:24